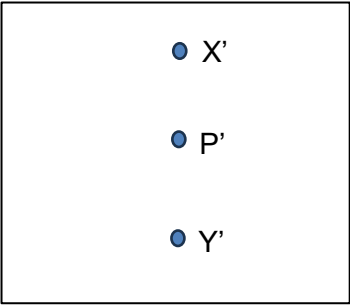


No.	Solution	Remarks
4(a)	1. Waves are coherent. 2. Waves have equal or approximately equal amplitudes. 3. Transverse waves must be either unpolarised or polarised in one plane.	[2] for any two correct conditions
4(b)(i)	$x = \frac{\lambda D}{a}$ $\lambda = \frac{ax}{D}$ $= \frac{0.10 \times 10^{-3} (0.90 \times 10^{-2})}{1.50}$ $= 6.0 \times 10^{-7} \text{ m}$	[1] for correct substitution [1] for correct answer
4(b)(ii)	Fringe separation remains unchanged. Dark fringes (or minima) appear less dark than before. Bright fringes (or maxima) appear less bright than before. or Contrast between the bright and dark fringes is less than distinct than before.	[1] [1]
4(c)(i)	$\tan \theta = \frac{38}{165}$ $\theta = 12.97^\circ$ $d \sin \theta = n\lambda$ $d = \frac{1(632 \times 10^{-9})}{\sin 12.97^\circ} = 2.82 \times 10^{-6} \text{ m}$ number of lines per metre $= \frac{1}{d}$ $= \frac{1}{2.82 \times 10^{-6}}$ $= 3.6 \times 10^5 \text{ m}^{-1}$	[1] for correct angle [1] for correct d [1] for correct answer
4(c)(ii)		[1] for correct positions of X' and Y' [1] for correct position of P'

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No.	Solution	Remarks